

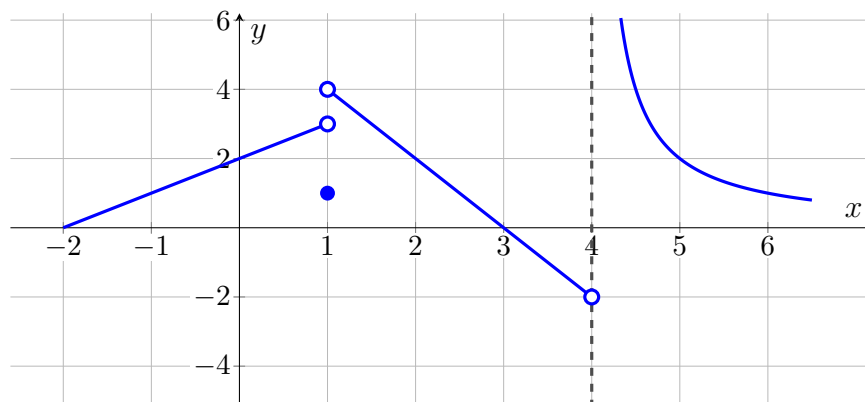
Reading a Limit Three Ways

Limit evidence from graphs, from tables, and from algebra

Directions:

- A limit can be read off a graph, estimated from a table, or computed by algebra. The three must agree — and when they seem not to, the algebra is what settles it.
- Problems 1, 2, 5 and 8 all refer to **Graph A** below. The rest carry their own table or expression.
- Write $+\infty$, $-\infty$, or DNE where a finite value does not exist, and say which.

Graph A — the graph of $y = f(x)$ used by problems 1, 2, 5 and 8. Open circles mark points not on the graph; the solid dot at $(1, 1)$ is the actual value $f(1)$. The dashed line $x = 4$ is a vertical asymptote.



1. Graph A. Read the two one-sided limits at $x = 1$.

(a) $\lim_{x \rightarrow 1^-} f(x) = \underline{\hspace{2cm}}$ (b) $\lim_{x \rightarrow 1^+} f(x) = \underline{\hspace{2cm}}$

2. Graph A. The graph passes through $x = 3$ with no break and no hole. Read $\lim_{x \rightarrow 3} f(x)$.

Answer: _____

3. The table lists values of $g(x) = \frac{x^3 - 27}{x - 3}$ near $x = 3$.

x	2.9	2.99	3.01	3.1
$g(x)$	26.11	26.9101	27.0901	27.91

Use the table to predict $\lim_{x \rightarrow 3} g(x)$, then confirm the prediction by factoring $x^3 - 27$ and cancelling.

Answer: _____

4. Compute algebraically — no table, no graph:

$$\lim_{x \rightarrow 0} \frac{\sqrt{x + 16} - 4}{x}$$

Answer: _____

- 5. Graph A.** Explain whether $\lim_{x \rightarrow 1} f(x)$ exists, using the two values you read in problem 1. Then read $f(1)$ from the solid dot and say how that third number compares with the other two.

Answer: _____

- 6.** The table lists values of $q(x) = \frac{|x - 5|}{x - 5}$ near $x = 5$.

x	4.9	4.99	5.01	5.1
$q(x)$	-1	-1	1	1

- (a) $\lim_{x \rightarrow 5^-} q(x) =$ _____ (b) $\lim_{x \rightarrow 5^+} q(x) =$ _____

- 7.** Compute algebraically:

$$\lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 - 4}$$

Answer: _____

8. Graph A. The dashed line $x = 4$ is a vertical asymptote, and the branch to its left ends at an open circle.

(a) $\lim_{x \rightarrow 4^-} f(x) = \underline{\hspace{2cm}}$ (b) $\lim_{x \rightarrow 4^+} f(x) = \underline{\hspace{2cm}}$

9. Compute algebraically:

$$\lim_{x \rightarrow 4} \frac{\sqrt{x+5} - 3}{x - 4}$$

Answer: $\underline{\hspace{2cm}}$

10. Synthesis challenge. Let $r(x) = \frac{x^2 - 9}{|x - 3|}$. Here is a table of values near $x = 3$.

x	2.9	2.99	3.01	3.1
$r(x)$	-5.9	-5.99	6.01	6.1

(a) $\lim_{x \rightarrow 3^-} r(x) = \underline{\hspace{2cm}}$ (b) $\lim_{x \rightarrow 3^+} r(x) = \underline{\hspace{2cm}}$

Confirm both values by splitting $|x - 3|$ into its two cases and cancelling — the table predicts, the algebra proves.

(c) Sketch what the graph of r must look like just left and just right of $x = 3$, and state whether $\lim_{x \rightarrow 3} r(x)$ exists.